

A smart mushroom incubation system for enhanced yield and sustainability in
small scale farming

RP 25-26J-264

Project Proposal Report

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Signature

Date

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Signature of the Supervisor

Date

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Signature of the Co-Supervisor

Date

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Abstract

Small-scale oyster mushroom cultivators in Sri Lanka often struggle with inconsistent yield and quality due to manual monitoring and control of key environmental parameters such as humidity, temperature, carbon dioxide (CO₂), and light. These variations not only affect fruiting initiation but also increase the risk of contamination and resource inefficiency [1]. To address this issue, this research proposes the design and implementation of an IoT-enabled, machine learning–based predictive mushroom fruiting system. The system integrates real-time environmental sensing with data-driven forecasting to determine optimal fruiting readiness windows [3]. By leveraging historical environmental data and actual fruiting events, the proposed model can accurately predict upcoming fruiting phases [4].

Based on these predictions, the system delivers proactive alerts to cultivators through a farmer-friendly interface. This predictive and automated approach moves beyond traditional timer- or rule-based methods, enabling higher yield consistency, improved resource utilization, and enhanced sustainability for small-scale oyster mushroom farming in Sri Lanka [5]. In addition, the solution promotes low-cost smart farming practices, reduces manual workload, and supports early detection of unfavorable conditions, making it practical and scalable for local farmers [6].

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Introduction

Mushroom cultivation is a rapidly growing sector in agriculture, valued for its nutritional benefits, medicinal properties, and economic potential [1][7]. Successful mushroom farming, however, requires precise environmental control to ensure optimal growth and timely fruiting [8]. Factors such as carbon dioxide (CO₂) concentration, temperature, humidity, and time play a critical role in determining the health and yield of mushrooms. Traditionally, small-scale farmers rely on manual observation to monitor these parameters, a practice that often leads to inconsistent yields, delayed harvesting, and suboptimal use of resources [1][3][10].

Recent advances in technology, particularly in the fields of the Internet of Things (IoT) and machine learning (ML), offer new opportunities to enhance mushroom cultivation [4][9]. IoT sensors can collect real-time data on environmental factors, enabling more precise monitoring of growth conditions [12]. Machine learning algorithms can then analyze this data to detect patterns and predict outcomes such as fruiting readiness, which is otherwise difficult to determine accurately using conventional methods [9][14].

State-of-the-art approaches in mushroom cultivation monitoring often focus on individual parameters. Some systems measure temperature and humidity, while others track CO₂ levels to a limited extent [7][10]. Automated environmental control systems exist for large-scale operations but are expensive and not suitable for small farmers [11]. Other research has applied predictive models, yet these are often based on limited datasets or controlled laboratory conditions, limiting their practical applicability in real-world farms [11][15].

Several solutions have been proposed to address these challenges. Some commercial systems provide IoT-based monitoring but lack predictive capabilities, leaving farmers to make decisions manually [12]. Other research projects have attempted to use computer vision or machine learning to estimate mushroom growth stages, but they often require costly hardware or are limited to specific mushroom species [13]. For example, one study introduced an autonomous mobile platform integrating ROS architecture, environmental sensors, and YOLOv4-based vision to identify and count king oyster mushrooms, thereby improving inspection efficiency in mushroom farms [2][13].

The system proposed in this project addresses these gaps by integrating IoT-based real-time monitoring of CO₂, temperature, humidity, and time with a machine learning model designed to predict mushroom fruiting readiness. Unlike existing methods, this approach provides small-scale farmers with an affordable, data-driven decision support tool that enhances yield prediction without the need for expensive automation [5][14]. By delivering timely alerts via a user-friendly interface, the system empowers farmers to optimize harvesting schedules and improve sustainability in mushroom cultivation [6].

Background & Literature Survey

Mushroom cultivation is a vital component of modern agriculture, valued for its nutritional content, medicinal properties, and economic potential, yet it remains highly sensitive to environmental conditions such as temperature, humidity, and carbon dioxide (CO₂) concentration [7][8]. These factors play a crucial role in determining the growth rate, quality, and timing of mushroom fruiting, and even minor deviations can result in reduced yields or delayed harvesting [8].

Traditionally, mushroom farmers rely on manual observation and experience to monitor these parameters, a method that is time-consuming, inconsistent, and prone to human error [10]. In recent years, the integration of IoT technologies in agriculture has enabled real-time monitoring of environmental conditions, allowing farmers to collect continuous data on key parameters [3][12]. Parallel advancements in machine learning have created opportunities to analyze such data, identify patterns, and make predictive assessments [4][9][14].

Most existing research and commercial systems focus on monitoring single parameters or operate in controlled laboratory settings, limiting their practicality for real-world farming [11]. Previous studies have attempted to classify mushroom growth stages using computer vision [13] or to automate environmental controls in large-scale farms, but these solutions are often expensive, complex, or not adaptable for smaller growers [15].

The novelty of this project lies in its integration of multiple IoT sensors with a predictive machine learning model that forecasts mushroom fruiting readiness, providing actionable insights based on real environmental data rather than simple observation or fixed thresholds [4][5]. By delivering timely alerts through a user-friendly interface, this system empowers farmers to make informed decisions regarding harvesting schedules, resource allocation, and overall farm management [6].

Research Gap

Research A - Research A focuses on the design of an automated incubator system for oyster mushroom cultivation in Mongolia, where the growing season is limited due to extremely cold winters. The study developed a model that maintains essential parameters—soil moisture, air humidity, soil and air temperature, and CO₂ levels—over a 30-day cycle without human intervention. The system demonstrated the potential to create a favorable artificial environment for mushroom cultivation and sustain growth automatically. However, while it provides valuable insights into environmental control for mushroom incubation, it primarily emphasizes short-term cultivation in controlled conditions and does not incorporate predictive modeling or decision-support tools for

Research B - Research B presents the Smart Mushroom House (SMH) project in Malaysia, designed to automate shiitake mushroom cultivation using sensor technology with feedback systems to regulate climate conditions such as CO₂, humidity, temperature, and light. The system was motivated by the growing mushroom industry in Malaysia and the need to increase yields through modern farm technologies. The SMH introduced incubation and grow rooms to optimize the production cycle, particularly during the critical incubation stage, aiming to boost productivity and export potential. While effective for shiitake mushrooms and large-scale operations, the system requires significant facility upgrades and does not address the affordability and practicality issues faced by small-scale farmers.

Research C - Research C introduces an autonomous mobile IoT platform designed to automate inspection and monitoring in king oyster mushroom farms in Taiwan. The platform integrates ROS-based modules including navigation, robotic arm control, environmental sensing, and camera nodes. It collects real-time environmental data such as temperature, humidity, and CO₂, which are stored in a networked database and visualized using Grafana. Notifications are sent to farm managers through LINE Notify, while a thermal imaging sensor combined with a YOLO V4 neural network is used to estimate mushroom yield. This system significantly reduces the need for manual labor in hazardous farm environments and enhances productivity. However, its implementation requires sophisticated robotics and computing resources, making it more suitable for large-scale, high-tech farms rather than small or low-cost cultivation setups.

Research D - Research D presents the development of SENSEPACK, an IoT-based smart monitoring and control system for mushroom cultivation in Bangladesh. The system uses sensors to measure temperature, humidity, CO₂ levels, and light intensity, with data accessible through a dedicated Android/web application. It allows both manual and automated control of external devices such as water pumps, exhaust fans, air conditioners, and light sources to maintain optimal growing conditions. The system is designed to support multiple mushroom species including oyster, milky, button, and shiitake, enabling year-round cultivation. While SENSEPACK demonstrates a practical and scalable solution for improving mushroom yields, it mainly focuses on environmental monitoring and control, with less emphasis on predictive analytics, automation of cultivation processes, or integration with AI-driven decision-making.

Application Reference	Real-time Environmental Monitoring (Temp, Humidity, CO ₂ , Light)	Automated Environmental Control	Mobile/Remote Monitoring	Machine Learning / Predictive Analytics	Fruiting Readiness Forecast	Small-scale, Cost-effective Solution
Research A (Automated Incubator – Mongolia)	✓	✓	X	X	X	X
Research B (Smart Mushroom House – Malaysia)	✓	✓	X	X	X	X
Research C (Autonomous IoT Robot – Taiwan)	✓	✓	✓	✓	X	X
Research D (SENSEPACK – Bangladesh)	✓	✓	✓	X	X	✓
Proposed System	✓	✓	✓	✓	✓	✓

Research Problem

Mushroom cultivation is an increasingly popular agricultural practice due to its high nutritional value, economic potential, and relatively low land requirements [7]. In Sri Lanka, oyster mushrooms are commonly cultivated by small-scale farmers as an income source. However, one of the major challenges these cultivators face is ensuring consistent yield and quality, as mushrooms are highly sensitive to environmental variations [8][10]. The fruiting process depends on maintaining specific levels of humidity, temperature, carbon dioxide (CO₂), and light within the growing environment [3][7]. Even slight deviations in these parameters can delay fruiting, reduce yield, and increase the risk of contamination, leading to financial losses for farmers [11].

At present, many small-scale cultivators rely on manual monitoring and experience-based adjustments to control their growing conditions [10]. This approach is highly labor-intensive, time-consuming, and prone to human error. Farmers often lack the tools to track subtle changes in CO₂, humidity, or temperature in real time, making it difficult to anticipate fruiting readiness [3][12].

Existing automated systems for mushroom farming typically focus on real-time monitoring or rule-based environmental control [10]. While these approaches can automate certain tasks such as misting or ventilation, they do not provide predictive insights into when mushrooms will be ready for fruiting [11][14]. Without predictive support, farmers continue to face uncertainty in planning harvesting, scheduling labor, and optimizing environmental resource usage. Moreover, many of the available advanced solutions are designed for large-scale commercial farms, making them too costly and technologically complex for small-scale Sri Lankan farmers [6][15].

Therefore, there exists a critical research problem: how to design a low-cost, IoT-enabled, machine learning-based predictive system that not only monitors real-time environmental conditions but also forecasts mushroom fruiting readiness [4][5]. Such a system would empower small-scale farmers with data-driven decision support, allowing them to act in advance rather than react after changes occur [14].

Objectives

Main Objectives

To design and implement a predictive mushroom fruiting system that combines real-time IoT environmental sensing with machine learning to forecast fruiting readiness, thereby improving yield consistency, sustainability, and decision-making for small-scale mushroom farmers in Sri Lanka [3][4][5].

Specific Objectives

- To develop an IoT-based monitoring framework that continuously records CO₂, humidity, temperature, and time from the cultivation environment [3][12].
- To preprocess and store historical sensor data alongside actual fruiting timelines for model training [4][9].
- To design and train a machine learning model capable of predicting optimal mushroom fruiting readiness windows [5][14].
- To integrate predictive insights into a user-friendly interface that provides real-time monitoring, visualizations, and alerts to farmers [6].
- To evaluate the system in terms of prediction accuracy, usability, cost-effectiveness, and scalability for small-scale mushroom farms [6][15].

Methodology

The proposed research adopts a design and implementation methodology that integrates real-time environmental sensing, data preprocessing, machine learning model development, and system integration for automated mushroom fruiting prediction [3][4][5]. The process begins with data collection, where IoT-enabled environmental sensors will be installed in the mushroom cultivation environment to continuously measure temperature, humidity, carbon dioxide (CO₂), and time. These parameters are chosen as they directly influence mushroom growth and fruiting readiness. Data will be collected across multiple cultivation cycles to capture a diverse range of environmental conditions, including both optimal growth phases and disturbances caused by variations in humidity, temperature, or CO₂ [7][8][12].

Following data collection, preprocessing will be performed to clean and prepare the dataset for machine learning. The raw sensor readings will undergo noise reduction, handling of missing values, and removal of inconsistent records. These cleaned data points will then be aligned with manually recorded mushroom fruiting events, creating a labeled dataset suitable for supervised learning. Feature engineering techniques such as normalization, scaling, and time-series windowing will be applied to improve the robustness and predictive power of the model [4][9][14].

Once the dataset has been prepared, the machine learning model will be developed and evaluated. Several algorithms will be tested, including decision trees, random forests, and recurrent neural networks (RNN), as these are well-suited for handling time-series agricultural data [5][9]. The dataset will be split into training, validation, and testing subsets to avoid overfitting and ensure generalizability across different cultivation cycles. Model performance will be measured using evaluation metrics such as accuracy, precision, recall, and F1-score, while hyperparameter tuning will be conducted to further optimize predictive accuracy [14].

Finally, the trained model will be integrated into a microcontroller-based control system that interfaces with both the IoT sensors and the mobile application. The predictive model's output will be used to generate proactive alerts for farmers regarding fruiting readiness. A user interface will be developed using Flutter and Firebase, providing cultivators with access to real-time monitoring, historical trend visualization, and predictive insights through a mobile dashboard [6][12]. This integration ensures that the system remains cost-effective, scalable, and farmer-friendly, offering a practical solution for small-scale mushroom growers in Sri Lanka. By embedding predictive analytics into the cultivation process, the methodology shifts traditional farming practices from reactive to proactive decision-making, thereby enhancing yield consistency, reducing manual workload, and promoting sustainable resource utilization [6][15].

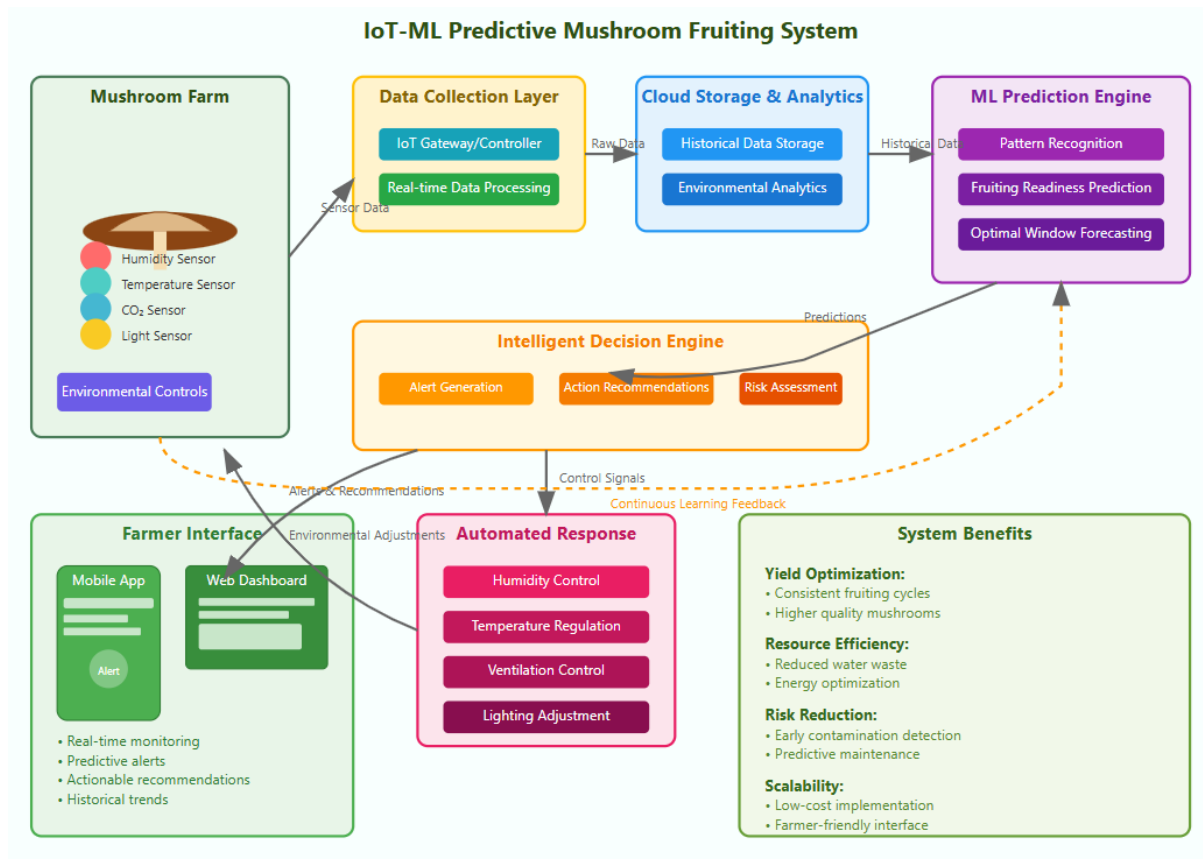


Figure 1: Predictive fruiting module using environmental sensing and machine learning

Project requirements

1. Functional requirements

- **Real-time Environmental Monitoring**

The system shall continuously monitor temperature, humidity, and CO₂ levels inside the mushroom cultivation environment using IoT sensors.

The system should record and store sensor data with timestamps for further analysis.

- **Data Acquisition and Storage**

The system should collect data from all sensors and store it in a structured database for easy retrieval.

The system shall maintain historical records of environmental conditions to support trend analysis and model training.

- **Machine Learning Prediction**

The system shall analyze the collected environmental data using a machine learning model to predict mushroom fruiting readiness.

The system should update predictions periodically as new sensory data is collected.

- **Alert and Notification System**

The system should provide timely alerts to the farmer when mushrooms are approaching fruiting readiness.

Alerts should be delivered via a user-friendly interface (e.g., mobile app or web dashboard).

- **User Interface**

The system shall provide an intuitive dashboard showing real-time environmental parameters, historical trends, and fruiting predictions.

The interface shall allow farmers to view predictions for multiple cultivation units if needed.

- **Manual Decision Support**

The system shall support farmers in making manual decisions regarding harvesting, without automating environmental controls.

The system shall allow users to acknowledge or dismiss alerts and track previous harvesting actions.

- **Data Export and Reporting**

The system shall allow exporting historical sensor data and predictions in standard formats (e.g., CSV or Excel) for analysis.

The system shall generate summary reports on environmental conditions and predicted fruiting timelines.

2. Non-Functional Requirements

- **Performance** – The system shall process sensor data and update predictions in near real-time.
- **Reliability** – The system shall operate continuously with minimal downtime and preserve data during temporary failures.
- **Scalability** – The system shall support additional cultivation units, farms, or sensors without major redesign.
- **Usability** – The interface shall be intuitive and understandable for users with limited technical knowledge.
- **Maintainability** – The system shall have modular components for easy updates, maintenance, or bug fixes.
- **Security** – The system shall protect all data and restrict access to authorized users.
- **Portability** – The interface shall be accessible via web and mobile devices, and the system deployable on different hardware.
- **Data Integrity** – The system shall ensure sensor readings and predictions are accurate and reliable.
- **Environmental Tolerance** – IoT sensors shall function reliably in high humidity and variable temperature conditions.

- **Cost-Effectiveness** – The system shall be affordable for small-scale farmers, avoiding expensive automation.

3. User requirements

Real-Time Monitoring – Users shall be able to view real-time data on CO₂ levels, temperature, and humidity in their mushroom cultivation environment.

Prediction of Fruiting – Users shall receive notifications or alerts when mushrooms are approaching fruiting readiness.

Data Visualization – Users shall see environmental data trends and predictions in graphs or charts that are easy to understand.

Historical Data Access – Users shall be able to review past environmental data and fruiting cycles for analysis and decision-making.

User-Friendly Interface – The system shall have a simple and intuitive interface accessible on web or mobile devices.

Customizable Alerts – Users shall be able to configure alert thresholds and notification preferences.

Decision Support – The system shall assist users in making informed harvesting decisions without automating the cultivation process.

Secure Access – Users shall have secure login credentials, ensuring that only authorized persons can access their data.

Multi-Unit Support – Users shall be able to monitor multiple cultivation units or rooms simultaneously.

Affordable and Practical – The system shall be cost-effective and suitable for small-scale farmers, requiring minimal technical expertise to operate.

4. System requirements

The purpose of software requirements is to define the software resources that must be enforced on a system for the proposed predictive mushroom fruiting system to function properly. The software specification requirements for this proposed component are as follows:

Flutter – To develop a cross-platform mobile application for both Android and iOS, allowing farmers to monitor environmental conditions and receive predictive alerts.

Firebase – To handle real-time database management, authentication, cloud storage, and push notifications for user alerts regarding mushroom fruiting readiness.

Spring Boot – To implement the backend server for processing sensor data, managing the machine learning predictions, and providing RESTful APIs for the mobile application.

Python with Keras / TensorFlow – To develop and train machine learning models that predict mushroom fruiting readiness based on environmental sensor data.

Integration between Spring Boot and Firebase – To synchronize the backend data with the mobile application in real time and maintain historical records for trend analysis.

RESTful APIs – To facilitate communication between the mobile app and backend server for data retrieval, prediction updates, and user notifications.

Data visualization libraries (e.g., charts_flutter) – To display environmental trends and fruiting predictions in an intuitive dashboard on the mobile application.

Secure Authentication and Authorization – To ensure only authorized users can access the system data and predictions.

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